

Emmanuel Isaías Guízar Bayardo

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Profile

Neuroscience student specializing in neuroengineering; develops brain-computer interfaces (BCI), signal processing pipelines (EEG, EMG), and AI-assisted diagnostic tools. President of DUNNE at UNAM.

Education

UNAM, School of Medicine

B.S. in Neuroscience; GPA: 9.30/10.0

Mexico City, Mexico

August 2023 – June 2027 (expected)

UVEG: Virtual University of the State of Guanajuato

B.S. in Computer Systems Engineering; GPA: 9.80/10.0

Guanajuato, Mexico (online)

July 2025 – July 2028 (expected)

Research and Projects

Parkinson's Disease Detection from EEG

Subject-wise validation and transferability to Emotiv EPOC

Thesis, UNAM

June 2025 – Present

- Replicated the Aljalal et al. (2022) framework on *SanDiego* ($n = 31$) with *subject-wise* validation (GroupKFold) and 1,536 features per epoch via Wavelet Packet Decomposition; exposed information leakage by contrasting segment-wise ($\text{AUROC} \approx 0.99\text{--}1.00$) with subject-wise ($\approx 0.63\text{--}0.72$), retaining 94–99% of performance on the 14 Emotiv EPOC channels.
- Presented the findings in the poster *Honest validation reshapes EEG-based Parkinson's disease detection* at the 10th Neuroengineering Summer School (Camogli, Italy, 2026).

Neural Population Dynamics & Decoding of NHP Electrophysiology

Computational models on laboratory-provided primate recordings

IFC, UNAM (Prof. Rossi Pool)

August 2025 – June 2026

- Analyzed in Python in vivo primate electrophysiology (73 single-unit neurons and LFP) from S1, S2, and premotor/prefrontal (MPC, DPC) cortex: single-neuron coding (mutual information, ROC/AUROC, choice probability) and population methods (PCA, demixed-PCA, targeted dimensionality reduction, neural manifolds) to decode stimulus, category, decision, and movement, with LFP connectivity (Granger, spike-field). Data provided by the laboratory.

ConnMorph: Morphometry and Functional Connectivity

Structural VBM and functional networks (*rs-fMRI*) with FSL and Nipype

Academic project, UNAM

March 2026 – June 2026

- Implemented a VBM pipeline on 12 T1 images (BET, FAST, FLIRT, FNIRT to MNI152, Jacobian modulation, FWHM ≈ 8 mm; 18 ROIs) and reconstructed resting-state networks of 7 subjects via group ICA (MELODIC, 30 components), recovering 4 canonical networks through the Yeo atlas and Dice (visual 0.633); a reproducible end-to-end pipeline.

WristQuest: Serious Game for Neurorehabilitation

Lead researcher; real-time sEMG for pediatric USCP

LANR-IFC, UNAM

February 2026 – May 2026

- Designed as lead researcher a serious game controlled by real-time sEMG (4 channels; 20–450 Hz band-pass, 60 Hz notch, RMS envelope, MVC normalization); computed the pilot clinical trial ($n = 52$, Cohen's $d = 0.8$, 40 sessions), authored the 58-page protocol with 7 clinical annexes, and implemented a five-panel Python dashboard; advisor: M.Sc. Yoás S. Ramírez Graullera.

NeuroDAC: Academic Demonstrator of Neuroengineering

Modular EEG and BCI interface; Python

DUNNE, UNAM

May 2025 – March 2026

- Led the development of a modular platform acquiring 1-channel EEG (NeuroSky MindWave Mobile 2) with real-time visualization (Dash/Plotly) and two neurofeedback modules; presented to over 600 attendees across six forums (IV Neurosymposium, Universum, C3, BioHack CCM, Student Societies Fair, and TeleExpo 2026).

Academic Experience

10th Neuroengineering Summer School (Camogli, Italy)

Selected participant (one of 30 international researchers)

University of Genoa (DIBRIS)

June 2026

- Completed advanced sessions after competitive review; executed hands-on protocols with polymer scaffolds for 2D/3D in vitro neural networks, multi-electrode array (MEA) recordings, and advanced-microscopy image processing.

ORBIT: Tech-Based Venture Acceleration

Graduate (one of 19 projects); "Neuro Wear" project (ideation)

MIT and Tec de Monterrey

March 2026 – June 2026

Sinapsis Winter 2026: Teaching Assistant

EMG course; biomedical PCBs and 3D printing for biological models

ITESM, Mexico City

December 2025 – January 2026

IV Neurosymposium: Main author and workshop instructor

Poster on brain disorder diagnosis via EEG and ML

ENES Juriquilla, UNAM

July 2025 – November 2025

NeuroSummer 2025 + Scientific Communication (UCLA)

Intersemester courses (35 h) and published outreach article

UNAM / Academic Memories

June 2025 – August 2025

Calcium-Imaging Laboratory (Bargas Lab)

Training in ex vivo experimental techniques; Parkinson's mouse model

IFC, UNAM

August 2024 – May 2025

Leadership

University Division of Neuroengineering (DUNNE), UNAM

President

Mexico City, Mexico

February 2026 – Present

- Lead multidisciplinary teams in BCI and AI-assisted diagnostics; grew the community from 60+ to 100+ members (~70%) while managing a full-capacity curriculum.
- Coordinate collaborations with 3 schools, 3 institutes, companies (Odoo, UNIT Electronics), and student societies; represent DUNNE at academic and industry events (Talent Land, Baxter, BioHack CCM) and drove donations of 350+ meals to two hospitals.

University Division of Neuroengineering (DUNNE), UNAM

Head of Computational Development

Mexico City, Mexico

November 2024 – January 2026

- Led the technical development of NeuroDAC; implemented ML models and EEG signal processing in Python (wavelets, entropy), programmed two BCI-connected games, and automated recruitment in Odoo, sustaining 60% of current recruitment.

2nd National Forum: 50 Young Leaders in Digital Health

Selected youth voice; Transform Health México / Youth

Mexico City, Mexico

April 2026

Awards and Honors

- Full Board & Training Grant, University of Genoa / DIBRIS (Camogli, Italy, June 2026); 50 Young Leaders in Digital Health in Mexico (Transform Health, April 2026).
- 2nd place (of 30 teams), MeIA 2025 (voice control, ~91%) and MeIA 2023 (DNA classification, ~93%); Honorable Mention, COVID-19 Hackathon (2025); winning poster, Neuroverbena 2024; third in IB class (2022).

Certifications and Complementary Training

- **BCI & Neurotechnology Spring School 2026:** g.tec, 130 h; final grade 100/100; BR41N.IO Hackathon.
- **University Microcredential in Research Methodologies:** University of Cantabria; 75 h, 3.00 ECTS; Notable (SANFI Santander Scholarship, 2026).
- **Medical Informatics with AI: Hybrid Summer School** (UNAM IIMAS / TU Braunschweig / Hannover Medical School, 2025); **Cambridge C1 Advanced (CAE)**, **DSD II** (C1, 2022); **MOS Excel Associate**; **Mendix Rapid Developer** (#109511, 2026).

Technical Skills

Languages: Python (advanced), C, MATLAB, LaTeX

Signal processing, ML and decoding: NumPy, SciPy, pandas, scikit-learn, MNE-Python, PyWavelets, PyTorch, TensorFlow, Keras; FSL (MELODIC), Nipype, Yeo atlas; PCA/demixed-PCA, neural manifolds, population decoding, connectivity (Granger, spike-field)

Hardware and tools: Arduino, ESP32/ESP32-S3, commercial EEG headsets (NeuroSky), sEMG, biomedical PCBs, 3D printing, OpenCV, Git/GitHub

Spoken languages: Spanish (native), English (C1, Cambridge CAE), German (B2/C1, DSD II), Japanese (basic)